FOUR EQUIVALENT CHARACTERISATIONS OF FORMAL ETALENESS FOR ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3593-3605 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3607-3623, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally etale and formally unramified morphisms of algebraic spaces, universal first order thickenings, conormal sheaves and differentials; the chain of equivalences is retained.

Source reference. Original lemma definition-universal-thickening: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3128–3138. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-characterize-formally-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2810–2816. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S The following are equivalent:*

(1) *f is formally étale,*
(2) *f is formally unramified and the universal first order thickening of X over Y is equal to X,*
(3) *f is formally unramified and $\mathcal{C}_{X/Y} = 0$, and*
(4) *$\Omega_{X/Y} = 0$ and $\mathcal{C}_{X/Y} = 0$.*

Proof. Actually, the last assertion only make sense because $\Omega_{X/Y} = 0$ implies that $\mathcal{C}_{X/Y}$ is defined via Lemma lemma-characterize-formally-unramified and Definition definition-universal-thickening. This also makes it clear that (3) and (4) are equivalent.

Either of the assumptions (1), (2), and (3) imply that f is formally unramified. Hence we may assume f is formally unramified. The equivalence of (1), (2), and (3) follow from the universal property of the universal first order thickening X'

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of X over S and the fact that $X = X' \Leftrightarrow \mathcal{C}_{X/Y} = 0$ since after all by definition $\mathcal{C}_{X/Y} = \mathcal{C}_{X/X'}$ is the ideal sheaf of X in X'. $\square$